

BIOLOGY — GRADE 11 | REPRODUCTION IN ANIMALS

Quick Check

Lesson 3: Built for the Journey: Structure of Sperm and Ovum

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.1: Reproduction in Animals | Lesson 3

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Before looking at any slides, a learner predicts that the human ovum will be much larger than a sperm cell. Which reason best supports this prediction?

- A** The ovum must be big enough to be seen by the sperm
- B** The ovum has to swim a long distance, so it needs a big tail
- C** The ovum has twice as many chromosomes as the sperm, so it takes up more space
- D** The ovum stores food in its cytoplasm and has a protective layer, but it does not swim

Under a microscope, a learner sees that the midpiece of a sperm cell is packed with mitochondria. What is the function of this structure?

- A** It supplies the energy that the flagellum needs for swimming
- B** It releases enzymes that digest the outer layer of the ovum
- C** It protects the sperm from being damaged inside the female body
- D** It stores the chromosomes that will be passed to the offspring

A human sperm cell is about 50 micrometres long. A human ovum is about 0.1 mm across (1 mm = 1000 micrometres). About how many times wider is the ovum than the sperm is long?

- A** 5 times
- B** 200 times
- C** 20 times
- D** 2 times

During fertilisation, the acrosome at the head of the sperm releases enzymes. What do these enzymes do?

- A** They give the sperm energy to swim
- B** They digest the jelly coat (zona pellucida) around the ovum so the sperm can enter
- C** They make the ovum produce more food reserves
- D** They join the sperm nucleus to the ovum nucleus

Both the sperm and the ovum have haploid nuclei (half the normal number of chromosomes). Why is this important for fertilisation?

- A** Haploid nuclei let the ovum divide without a sperm
- B** Haploid nuclei make the sperm lighter so it can swim faster
- C** The fused nuclei give the zygote the full normal number of chromosomes
- D** Haploid nuclei mean the offspring will have only half the genes of the mother

A frog at the Lake Victoria wetlands releases thousands of eggs, but a cow usually gives birth to one calf. Which explanation best accounts for this difference?

- A** Cows cannot make more than one ovum because they have no jelly coat
- B** Frogs have more sperm than cows, so they must also make more eggs
- C** Frog eggs are fertilised in open water and have little protection, so many are lost, while a cow's fertilised egg is protected and nourished inside her body
- D** Frog eggs are bigger than cow eggs, so more of them are needed

A news report says that in IVF the baby grows in a test tube from start to birth. Which statement correctly describes IVF?

- A** Fertilisation happens in a laboratory dish, and the embryo is then placed in the womb of the mother or a surrogate to develop
- B** Fertilisation and the whole development of the baby happen in the laboratory
- C** The ovum develops into a baby in the dish without any sperm
- D** Fertilisation happens in the womb, and only the sperm is prepared in the laboratory

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).